

cDFT_electrolytes

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June 16, 2026

1 Introduction

The Helmholtz free energy functional $\mathcal{F}$ is given by:

$$\mathcal{F}[\{\rho_i\}, \vec{P}] = \mathcal{F}_{\text{ideal}}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{excess}}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{external}}[\{\rho_i\}, \vec{P}] \quad (1)$$

We can further decompose this functional into two parts:

$$\mathcal{F}[\{\rho_i\}, \vec{P}] = \mathcal{F}^{(0)}[\{\rho_i\}] + \mathcal{F}^{(1)}[\{\rho_i\}, \vec{P}], \text{ where} \quad (2)$$

$$\mathcal{F}^{(0)}[\{\rho_i\}] = \mathcal{F}_{\text{ideal}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{external}}^{(0)}[\{\rho_i\}] \text{ and} \quad (3)$$

$$\mathcal{F}^{(1)}[\{\rho_i\}, \vec{P}] = \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] + \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{external}}^{(1)}[\{\rho_i\}, \vec{P}] \quad (4)$$

Some of these contributions are standard:

$$\mathcal{F}_{\text{ideal}}^{(0)}[\{\rho_i\}] = \beta^{-1} \sum_i \int \rho_i(\vec{x}) \left(\ln \rho_i(\vec{x}) - 1 \right) d\vec{x} \quad (5)$$

$$\mathcal{F}_{\text{external}}^{(0)}[\{\rho_i\}] = \sum_i \int \rho_i(\vec{x}) V_{\text{ext},i}(\vec{x}) d\vec{x} \quad (6)$$

$$\mathcal{F}_{\text{ideal}}^{(1)}[\{\rho_i\}, \vec{P}] = \beta^{-1} \int \rho_0(\vec{x}) \left[\ln \left(\frac{\mathcal{L}^{-1}(\Omega(\vec{x}))}{\sinh(\mathcal{L}^{-1}(\Omega(\vec{x})))} \right) + \Omega(\vec{x}) \mathcal{L}^{-1}(\Omega(\vec{x})) \right] \quad (7)$$

Here, $\Omega(\vec{x}) = \frac{P(\vec{x})}{p\rho_0(\vec{x})}$ (where, $P(\vec{x}) = |\vec{P}(\vec{x})|$) and $\mathcal{L}(x) = \coth(x) - \frac{1}{x}$ is the Langevin function.

2 Update equation (potential balance)

2.1 The densities

For the i -th component, the update looks like:

$$\rho_i(\vec{x}) = \exp\left\{-\beta V_{\text{ext}}(\vec{x}) + \beta \mu_i + c_i^{(1)}(\vec{x})\right\} \quad (8)$$

where the one-body effective potential $c_i^{(1)}(\vec{x})$ are given by:

$$c_i^{(1)}(\vec{x}) = -\beta \frac{\delta}{\delta \rho_i} \left[\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] + \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] \right] \quad (9)$$

Now, we have

$$\mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] = \frac{1}{2} \int \rho_q(\vec{x}) \phi(\vec{x}) d\vec{x} \quad (10)$$

where $\rho_q(\vec{x}) = \sum_i Q_i \rho_i(\vec{x}) - \nabla \cdot \vec{P}(\vec{x})$ is the source charge distribution and $\phi(\vec{x})$ is the electrostatic potential generated by it using the Poisson equation:

$$\nabla^2 \phi(\vec{x}) = -\frac{\rho_q(\vec{x})}{\epsilon_0} \quad (11)$$

If we include the boundary conditions while computing the electrostatic potential and the corresponding electric field ($\vec{E}(\vec{x}) = -\nabla \phi(\vec{x})$), we can safely choose $\mathcal{F}_{\text{external}}^{(1)}[\{\rho_i\}, \vec{P}] = 0$.

This leads to:

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess}}^{(1)}[\{\rho_i\}, \vec{P}] = Q_i \phi(\vec{x}) \quad (12)$$

and furthermore, $\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}]$ can be decomposed into contributions arising due to various terms:

$$\mathcal{F}_{\text{excess}}^{(0)}[\{\rho_i\}] = \mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] + \mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] \text{ where,} \quad (13)$$

$$\mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] = \int \Phi_{\text{HS}}(\{n_\alpha(\vec{x})\}) d\vec{x} \quad (14)$$

$$\mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] = \frac{1}{2} \sum_{i,j} \int \rho_i(\vec{x}_1) \rho_j(\vec{x}_2) U_{\text{LJ}}^{i,j}(|\vec{x}_1 - \vec{x}_2|) \quad (15)$$

This leads to:

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess,HS}}^{(0)}[\{\rho_i\}] = \sum_\alpha \int \frac{\partial \Phi_{\text{HS}}}{\partial n_\alpha(\vec{x}_1)} \frac{\delta n_\alpha(\vec{x}_1)}{\delta \rho_i(\vec{x})} d\vec{x}_1 \quad (16)$$

$$\frac{\delta}{\delta \rho_i} \mathcal{F}_{\text{excess,LJ}}^{(0)}[\{\rho_i\}] = \sum_j \int \rho_j(\vec{x}_1) U_{\text{LJ}}^{i,j}(|\vec{x}_1 - \vec{x}|) d\vec{x}_1 \quad (17)$$

and finally, the contribution to the solvent density due to density-polarization coupling through the $\mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}]$ term :

$$\frac{\delta}{\delta \rho_0} \mathcal{F}_{\text{ideal}}^{(1)}[\rho_0, \vec{P}] = \ln \left(\frac{\mathcal{L}^{-1}(\Omega(\vec{x}))}{\sinh(\mathcal{L}^{-1}(\Omega(\vec{x})))} \right) \quad (18)$$

2.2 The polarization

$$\vec{P}(\vec{x}) = p\rho_0(\vec{x})\mathcal{L}\left(\beta p|\vec{E}(\vec{x})|\right)\hat{E}(\vec{x}) \quad (19)$$